The Impact of Natural Disasters on Risk Preferences, Subjective Beliefs, and Related Behavior: Evidence from Typhoon Ketsana

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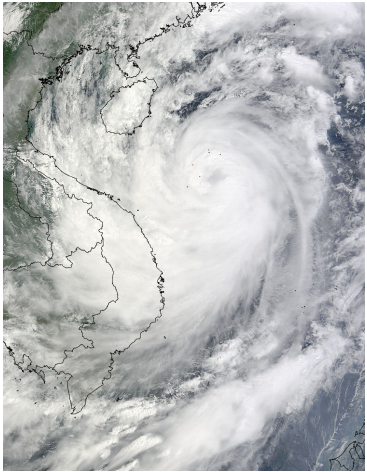
Risk Preferences Over Time

- A preference for risk refers to the tendency to engage in behaviors or activities that involve higher variance in returns
- Important to understand systematic changes in risk preferences as willingness to take risks can predict health outcomes, labor market/investment/migration decisions, addiction
- Many studies document the way risk preferences may be altered by negative shocks:
  - Financial crises (Dohmen et al., 2016; Guiso et al., 2018; Necker and Ziegelmeyer, 2016)
  - Violence/conflict (Callen et al., 2014; Jakiela and Ozier, 2019; Voors et al., 2012)
  - Natural disasters (Cameron and Shah, 2015; Cassar et al., 2017; Hanaoka et al., 2018)
- While financial shocks tend to make people more risk-averse, there is no general consensus on the direction in which violence/conflict or natural disasters alter risk preferences
- If a change in risk preferences does occur in the wake of a natural disaster, there is almost no evidence showing whether the effect is transitory or persistent

Typhoon Ketsana

- On September 29, 2009, Typhoon Ketsana made landfall in Central Vietnam
  - High wind speeds and torrential rainfall over three days leading to widespread flooding
- Ketsana was one of the most devastating storms in Vietnam since at least 1990
  - Category 2 typhoon—unusual for the country
  - 2.5 million people were affected by Ketsana, including 109,000 homeless, 860 injured, and 182 killed
  - Economic losses are estimated at USD 900 million
- **Did Ketsana alter individual risk preferences? In what direction? Did the effect persist over time?**
- **What were the effects on subjective beliefs about future shocks and risk-related behavior?**



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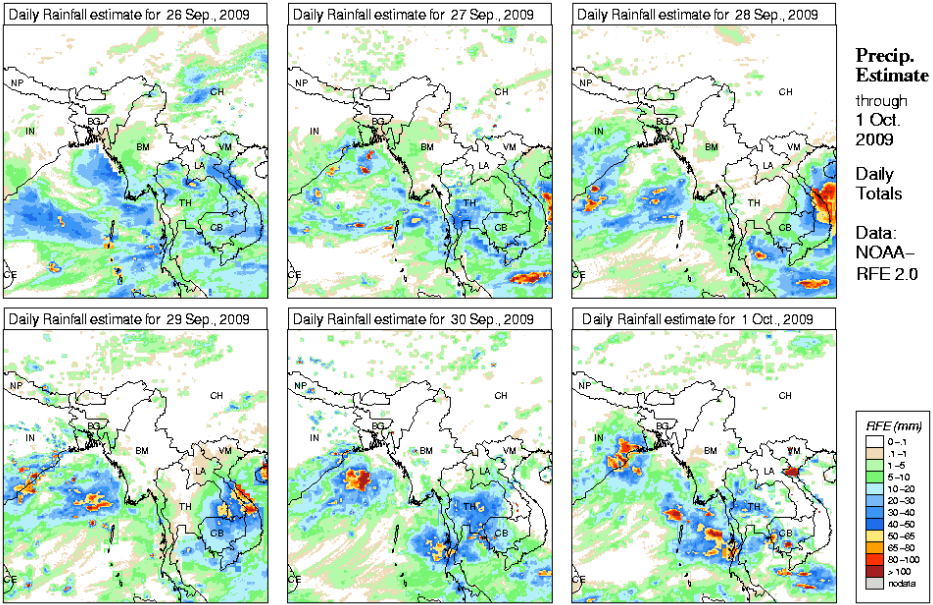


Precip. Estimate

through 1 Oct. 2009

Daily Totals

Data: NOAA–RFE 2.0

Data Sources

- NOAA's Rainfall Estimation Algorithm 2.0 (RFE 2.0)
  - Treatment variable—Rainfall: Excess rainfall within a 5 km radius of village during typhoon compared to normal times
- NASA's Moderate Resolution Imaging Spectroradiometer (MODIS)
  - Water coverage: Percentage of land area inundated within a 5 km radius of village during typhoon
- Thailand-Vietnam Socio-Economics Panel Survey (TVSEP)
  - Covers three Vietnam provinces that are relatively lower income and largely agricultural
  - Individual characteristics, socio-demographics, and self-reported shocks
  - Six waves: 2007, 2008, 2010, 2013, 2016, 2017
    - 2 waves pre- and 4 waves post-Ketsana
  - Main outcome variable: Risk aversion measured on a 0-to-10 scale via self-reports
    - Survey question: "Are you generally a person who is fully prepared to take risks or do you try to avoid taking risk?"
    - The lower the number a person reports, the more risk averse they are — 0 indicates "completely unwilling to take risks" while 10 indicates "fully prepared to take risks"

Measuring Risk Preferences

- Two main measurement tools have become well-established:
  1. Self-reports
     - ★ 11-point (0 to 10) Likert scale (Wagner et al., 2007)
  2. Incentivized experiments
     - ★ Choice between gambles (Binswanger, 1980; Eckel and Grossman, 2002)
     - ★ Risky investment task (Gneezy and Potters, 1997)
     - ★ Price lists (Holt and Laury, 2002)
- A couple of ways to think about how to validate risk preference measuring tools
  - Internal validity: Do different tools for measuring risk preferences map into underlying "risk preferences" and capture a description of the same individual?
  - External validity: Do risk measurement tools have predictive power for actual risk-related behavior?
- Survey measures of general risk-taking are typically highly correlated across different risk domains such as health, career, or financial matters, thus suggesting a single underlying trait determining risk
- In terms of external validity, self-reports seem to even outperform experiments (Dohmen et al., 2011)

Data Caveats

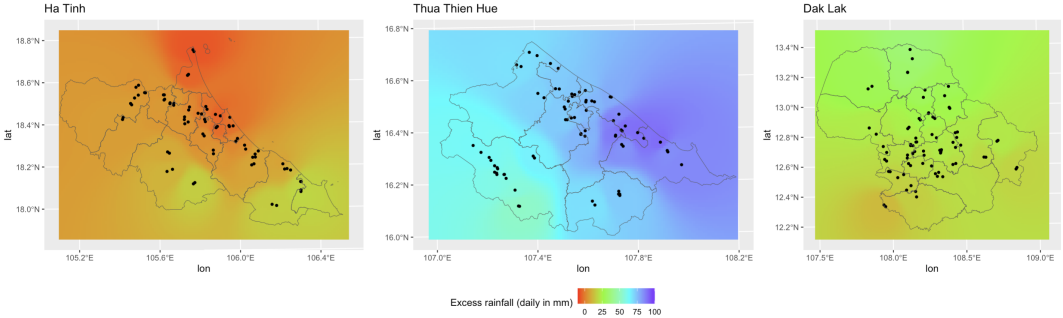
- The 2007 TVSEP wave does not collect risk preferences data, preventing us from testing for pre-trends on our dependent variable directly
  - We assess a few related variables for pre-trends instead
    - Total daily payments in insurance premiums is our preferred one
- We only use households where the respondent was the head of household and was the same respondent across 2007, 2008, 2010, and 2013 waves
  - Limiting sample to the head of household assuming this individual makes the household's financial decisions (pre-trend variables) according to their own risk preferences
- Because some baseline characteristics differ slightly across the rainfall distribution, we use entropy balancing to reweight observations.

Summary Statistics at Baseline (2008)

	count	mean	sd	min	max
Risk Preference					
Risk Preference	572	3.44	3.04	0.00	10.00
Individual Characteristics					
Household Size	572	4.77	2.12	1.00	14.00
Daily Expenditure (per capita, USD)	572	2.04	1.61	0.11	24.03
Daily Insurance Premiums (household, USD)	572	0.21	1.10	0.00	21.43
Age	572	51.67	13.68	22.00	91.00
Gender (1 = Male)	572	0.74	0.44	0.00	1.00
Self-Employed	571	0.81	0.39	0.00	1.00
Subjective Beliefs on Occurrence of Shocks					
Floods	562	1.87	2.24	0.00	6.00
Storms	566	2.30	2.43	0.00	6.00
House Damage	559	0.33	0.76	0.00	5.00
Household Shocks since Last Wave					
Death	572	0.03	0.18	0.00	2.00
Accident	572	0.02	0.15	0.00	1.00
Drought	572	0.10	0.35	0.00	3.00
Pest/Livestock Disease	572	0.30	0.51	0.00	2.00
Storm	572	0.16	0.37	0.00	1.00
House Damage	572	0.02	0.15	0.00	1.00
Treatment Variable					
Excess Rainfall (mm)	572	31.23	29.76	-6.21	90.23
Weather Benchmark					
Flooding Propensity (%)	572	0.15	0.11	0.02	0.60

Treatment Variable

Excess rainfall within a 5 km radius of village (average daily rainfall during the days of the typhoon minus average daily rainfall in days right before and right after the typhoon)



Identification Strategy

We use a difference-in-differences model with a continuous treatment variable exploiting variation in the intensity of the typhoon:

$$Y_{ivt} = Post_t + \beta T_v \times Post_t + \rho P_v \times Post_t + \gamma X_{ivt} + \delta Z_{ivt} + \pi_i + \varepsilon_{ivt},$$

where:

- i indexes the individual, v the village, t time ($t = 2008$, 2010, or 2013)
- Y_{ivt} is the self-reported risk preferences score for the main results
- $Post_t$ is a dummy that turns on when $t = 2010$ for short-term effects or $t = 2013$ for long-term effects
- T_v is a measure of excess rainfall within a 5 km radius of village v
- P_v is the water coverage within a 5 km radius of village v during normal times
- X_{ivt} is a vector of time-varying socio-demographic characteristics
- Z_{ivt} is a vector of controls for life changes and other shocks that may change risk preferences (Kettlewell, 2019)
- π_i is individual-level fixed effects

Baseline Balance Across Terciles of Excess Rainfall

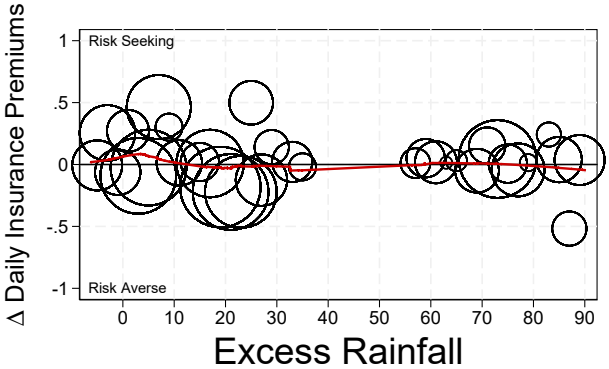
	Tercile 1 ($N = 189$)		Tercile 2 ($N = 195$)		Tercile 3 ($N = 188$)		T-test P-values		
Variables	Mean	(SE)	Mean	(SE)	Mean	(SE)	(1) − (2)	(1) − (3)	(2) − (3)
Risk Preference									
Risk Preference	2.32	(0.18)	3.06	(0.20)	4.96	(0.23)	0.01***	0.00***	0.00***
Individual Characteristics									
Household Size	5.08	(0.16)	4.64	(0.13)	4.59	(0.17)	0.03**	0.03**	0.81
Daily Expenditure (USD)	2.18	(0.15)	1.94	(0.10)	2.01	(0.10)	0.17	0.35	0.60
Assets (Log)	7.91	(0.09)	7.83	(0.09)	7.51	(0.08)	0.51	0.00***	0.01**
Insurance Premium (Daily)	0.35	(0.13)	0.16	(0.04)	0.12	(0.03)	0.15	0.08*	0.47
Age	48.46	(0.92)	51.47	(0.95)	55.10	(1.05)	0.02**	0.00***	0.01**
Gender (1 = Male)	0.76	(0.03)	0.73	(0.03)	0.72	(0.03)	0.45	0.33	0.83
Self-Employed	0.81	(0.03)	0.83	(0.03)	0.79	(0.03)	0.68	0.49	0.27
Household Shocks since Last Wave									
Death	0.02	(0.01)	0.03	(0.01)	0.05	(0.02)	0.39	0.08*	0.42
Accident	0.02	(0.01)	0.04	(0.01)	0.01	(0.01)	0.14	0.66	0.06*
Drought	0.21	(0.04)	0.08	(0.02)	0.03	(0.01)	0.00***	0.00***	0.05*
Pest/Livestock Disease	0.31	(0.04)	0.35	(0.04)	0.24	(0.03)	0.45	0.21	0.04**
Storm	0.10	(0.02)	0.19	(0.03)	0.21	(0.03)	0.01***	0.00***	0.67
House Damage	0.02	(0.01)	0.03	(0.01)	0.02	(0.01)	0.77	0.99	0.78
Subjective Beliefs about Occurrence of Shocks									
Floods	1.31	(0.14)	1.98	(0.18)	2.34	(0.16)	0.00***	0.00***	0.14
Storms	0.89	(0.14)	2.75	(0.19)	3.22	(0.15)	0.00***	0.00***	0.05**
House Damage	0.27	(0.04)	0.42	(0.07)	0.29	(0.05)	0.07*	0.67	0.15

Note: The first three columns report means with standard errors in brackets for households in the lowest, middle, and highest terciles of excess rainfall exposure, respectively. The subjective beliefs section records the number of times the respondent anticipates facing a given shock in the next five years.

Addressing Baseline Imbalance: Entropy Balancing

- Baseline values differ across terciles of excess rainfall for some covariates (e.g., household size, household head age, exposure prior shocks), so higher rainfall exposure is correlated with some pre-existing characteristics
- To ensure these differences do not confound our estimates, we reweight the sample using entropy balancing (Hainmueller, 2012)
  - This method constructs weights for the observations so that baseline covariates are exactly balanced across the continuous treatment variable, while avoiding model dependence required by other approaches
- As we will see, reweighted results remain very similar, strengthening our confidence that the estimated effects reflect typhoon exposure, and not baseline differences

Pre-Trends: No apparent relationship between excess rainfall during the 2009 typhoon and 2007–2008 change in insurance premiums



Note: Change in the daily insurance premiums (USD) between 2007 and 2008 on the *y*-axis. Excess rainfall in millimeters on the *x*-axis is calculated as daily average rainfall during landfall minus daily average rainfall before and after landfall. Individuals are classified into 1-millimeter bins according to the exposure to excess rainfall during the typhoon. The size of each circle represents the number of individuals in the corresponding bin. We plot a *lowess* curve with a bandwidth of 0.8.

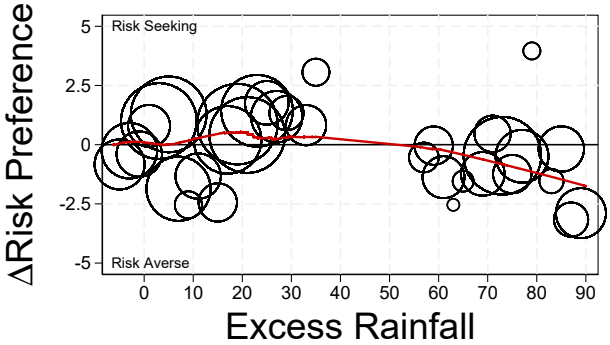
Pre-Trends: Daily Insurance Premiums

Pre-Treatment Trends: Changes in Daily Insurance Premium (2007–2008)

	(1)	(2)
Excess Rainfall	0.016	0.026
	(0.036)	(0.039)
Observations	1144	1144
Individual FE	✓	✓
Controls & Reweighting		✓

Note: The dependent variable is the total amount of insurance premiums paid by the household, calculated at the daily level. We report the difference-in-differences coefficient, i.e., the coefficient before the treatment interacted with a dummy for the 2008 wave. Controls include expenditure, assets, age, change in respondent's marital status, household size, exogenous shocks to the household including accidents, drought, pest and livestock diseases, crime, and death of a household member, and historical flooding propensity. Flooding propensity refers to the historical average flooding for the same time window around the typhoon in 2001–2008. Entropy balancing weights are applied, ensuring covariate balance conditional on rainfall intensity. Standard errors are in parentheses and clustered at the subdistrict level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Individuals exposed to higher excess rainfall during the 2009 typhoon seemingly become more risk averse over 2008–2010



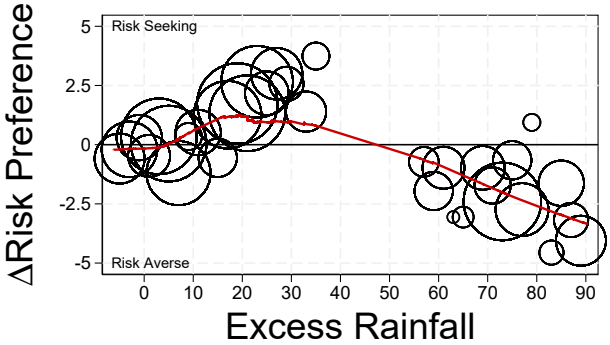
Note: Change in self-reported risk preferences (measured on a 0-to-10 scale) between 2008 and 2010 on the y-axis. Excess rainfall in millimeters on the x-axis is calculated as daily average rainfall during landfall minus daily average rainfall before and after landfall. Individuals are classified into 2-millimeter bins according to the exposure to excess rainfall during the typhoon. The size of each circle represents the number of individuals in the corresponding bin. We plot a lowess curve with a bandwidth of 0.8.

Results: Short-Term Changes in Risk Preferences (2008–2010)

	(1)	(2)
Excess Rainfall	-0.160***	-0.112**
	(0.057)	(0.056)
Observations	1144	1144
Individual FE	✓	✓
Controls & Reweighting		✓

Note: The dependent variable is the respondent's stated risk preference. We report the difference-in-differences coefficient, i.e., the coefficient before the treatment interacted with a dummy for the 2008 wave. Controls include expenditure, assets, age, change in respondent's marital status, household size, exogenous shocks to the household including accidents, drought, pest and livestock diseases, crime, and death of a household member, and historical flooding propensity. Flooding propensity refers to the historical average flooding for the same time window around the typhoon in 2001–2008. Entropy balancing weights are applied in column (2), ensuring covariate balance conditional on rainfall intensity. Standard errors are in parentheses and clustered at the subdistrict level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Individuals exposed to higher excess rainfall during the 2009 typhoon seemingly become more risk averse over 2008–2013



Note: Change in self-reported risk preferences (measured on a 0-to-10 scale) between 2008 and 2013 on the *y*-axis. Excess rainfall in millimeters on the *x*-axis is calculated as daily average rainfall during landfall minus daily average rainfall before and after landfall. Individuals are classified into 2-millimeter bins according to the exposure to excess rainfall during the typhoon. The size of each circle represents the number of individuals in the corresponding bin. We plot a *lowess* curve with a bandwidth of 0.8.

Results: Long-Term Changes in Risk Preferences (2008–2013)

	(1)	(2)
Excess Rainfall	-0.185***	-0.140***
	(0.033)	(0.032)
Observations	1144	1144
Individual FE	✓	✓
Controls & Reweighting		✓

Note: The dependent variable is the respondent's stated risk preference. We report the difference-in-differences coefficient, i.e., the coefficient before the treatment interacted with a dummy for the 2008 wave. Controls include expenditure, assets, age, change in respondent's marital status, household size, exogenous shocks to the household including accidents, drought, pest and livestock diseases, crime, and death of a household member, and historical flooding propensity. Flooding propensity refers to the historical average flooding for the same time window around the typhoon in 2001–2008. Entropy balancing weights are applied, ensuring covariate balance conditional on rainfall intensity. Standard errors are in parentheses and clustered at the subdistrict level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Results: Subjective Beliefs about Occurrence of Storms

How often do you think a storm will occur in next five years?

	(1)	(2)
Panel A: Short-Term Effects (2008–2010)		
Excess Rainfall	0.170**	0.269***
	(0.082)	(0.084)
Observations	1136	1136
Panel B: Long-Term Effects (2008–2013)		
Excess Rainfall	-0.025	0.002
	(0.037)	(0.042)
Observations	1138	1138
Individual FE	✓	✓
Controls & Reweighting		✓

Note: The dependent variable is the expected number of storms in the next five years. We report the difference-in-differences coefficient, i.e., the coefficient before the treatment interacted with a dummy for the 2008 wave. Controls include expenditure, assets, age, change in respondent's marital status, household size, exogenous shocks to the household including accidents, drought, pest and livestock diseases, crime, and death of a household member, and historical flooding propensity. Flooding propensity refers to the historical average flooding for the same time window around the typhoon in 2001–2008. Entropy balancing weights are applied, ensuring covariate balance conditional on rainfall intensity. Standard errors are in parentheses and clustered at the subdistrict level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Results: Risk-Related Behaviors

	Insurance Premiums (1)	Drinking & Smoking Budget Share (2)	Gambling Budget Share (3)	1{Drinking & Smoking} (4)	1{Gambling} (5)
Panel A: Short-Term Effects (2008–2010)					
Excess Rainfall	0.014	-0.029*	-0.001	0.050	-0.261***
	(0.024)	(0.015)	(0.002)	(0.061)	(0.095)
Observations	1144	1144	1144	1144	1144
Panel B: Long-Term Effects (2008–2013)					
Excess Rainfall	0.028*	-0.006	0.002	-0.003	-0.064
	(0.015)	(0.010)	(0.001)	(0.030)	(0.054)
Observations	1108	1102	1102	1144	1144

Note: We report the difference-in-differences coefficient, i.e., the coefficient before the treatment interacted with a dummy for the 2010 wave (short-term) or 2013 wave (long-term). Estimation includes individual fixed effects. Controls include expenditure, assets, age, change in marital status of respondent, household size, exogenous shocks to the household including accidents, drought, pest and livestock diseases, crime, and death of a household member, and historical flooding propensity. Flooding propensity refers to the historical average flooding for the same time window around the typhoon in 2001–2008. Entropy balancing weights are applied, ensuring covariate balance conditional on rainfall intensity. Standard errors in parentheses and clustered at the subdistrict level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Mediation Analysis: Subjective Beliefs about Storms

How often do you think a storm will occur in next five years?

	(1)	(2)	(3)	(4)
Excess Rainfall	-0.160***	-0.169***	-0.112**	-0.115**
	(0.057)	(0.054)	(0.056)	(0.056)
Storms		0.076**		0.042
		(0.037)		(0.037)
Observations	1144	1128	1144	1128
Individual FE	✓	✓	✓	✓
Controls & Reweighting			✓	✓

Note: The dependent variable is the respondent's stated risk preference. The risk preference variable is a response to the question "Are you generally a person who is fully prepared to take risks or do you try to avoid taking risk?", where 0 indicates "completely unwilling to take risks" and 10 indicates "fully prepared to take risks". We report the difference-in-differences coefficient, i.e., the coefficient before the treatment interacted with a dummy for the 2008 wave. The storms variable is the respondent's expected frequency of storms that will occur in the next five years. Controls include expenditure, assets, age, change in respondent's marital status, household size, exogenous shocks to the household including accidents, drought, pest and livestock diseases, crime, and death of a household member, and historical flooding propensity. Flooding propensity refers to the historical average flooding for the same time window around the typhoon in 2001–2008. Entropy balancing weights are applied, ensuring covariate balance conditional on rainfall intensity. Standard errors are in parentheses and clustered at the subdistrict level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Mediation Analysis: Psychological Channel

Do you think your household will be better off in five years? (reverse-coded)

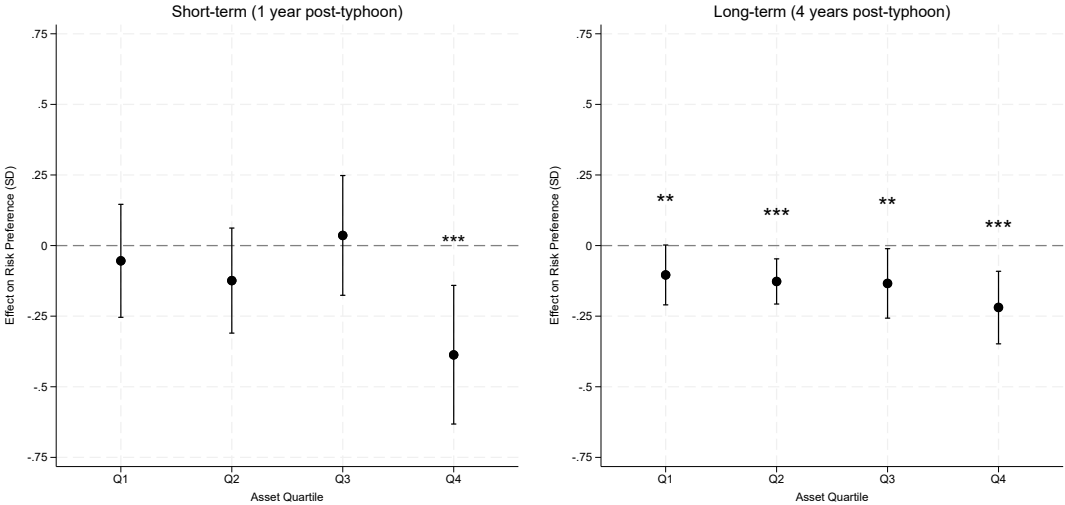
		Future Pessimism (1)
Panel A: Treatment Effect on Future Pessimisn		
Excess Rainfall		0.140***
		(0.038)
Observations		1048
	Risk Preference (1)	Risk Preference (2)
Panel B: Mediation Analysis		
Excess Rainfall	-0.112**	-0.090
	(0.058)	(0.060)
Future Pessimism		-0.149***
		(0.055)
Observations	1144	1048

Note: The dependent variable in Panel A is the respondent's subjective assessment of future well-being. The subjective well-being variable is a response to the question "Do you think your household will be better off in 5 years?", where 1 corresponds to "much better off" and 5 corresponds to "much worse off". Entropy balancing weights are applied, ensuring covariate balance conditional on rainfall intensity. Standard errors are in parentheses and clustered at the subdistrict level.

Mediation Analysis: Economic Channel

		Asset Losses (1)	Income Losses (2)	Extra Expenditure (3)
Panel A: Treatment Effect on Economic Outcomes				
Excess Rainfall		0.100**	0.116	0.238**
		(0.045)	(0.074)	(0.092)
Observations		1144	1144	1144
	Risk Preference (1)	Risk Preference (2)	Risk Preference (3)	Risk Preference (4)
Panel B: Mediation Analysis				
Excess Rainfall	-0.112**	-0.109*	-0.118**	-0.104*
	(0.058)	(0.056)	(0.055)	(0.055)
Asset Losses		-0.029		
		(0.022)		
Income Losses			0.051***	
			(0.014)	
Extra Expenditure				-0.033*
				(0.020)
Observations	1144	1144	1144	1144

Exploring Heterogeneity



Conclusion

- Natural disasters can lead to long-lasting impacts on risk attitudes
  - Individuals become more risk averse after the typhoon, and this effect persists even four years after the event
  - Belief updating is unlikely to explain these changes, as short-run expectations of storms increase, but controlling for beliefs does not attenuate the effect on risk preferences, and long-run expectations revert to baseline
  - Plausible mechanisms include increased pessimism about the future and short-run economic strain, with the psychological channel appearing stronger
  - A preliminary heterogeneity analysis shows that the increase in risk aversion is stronger among wealthier households
- Risk-related behavior adjusts over time, with individuals reducing their risky spending in the short run and increasing insurance purchases in the long run, consistent with a shift towards more caution
- Understanding these behavioral responses is important for designing post-disaster support and financial tools that align with how households adapt after shocks

Thank you! marcos.sugastti@ed.ac.uk

References I

Binswanger, H. P. (1980). Attitudes toward risk: Experimental measurement in rural india. *American Journal of Agricultural Economics*, 62(3):395–407.

Callen, M., Isaqzadeh, M., Long, J. D., and Sprenger, C. (2014). Violence and risk preference: Experimental evidence from afghanistan. *American Economic Review*, 104(1):123–148.

Cameron, L. and Shah, M. (2015). Risk-taking behavior in the wake of natural disasters. *Journal of human resources*, 50(2):484–515.

Cassar, A., Healy, A., and Von Kessler, C. (2017). Trust, risk, and time preferences after a natural disaster: experimental evidence from thailand. *World Development*, 94:90–105.

Dohmen, T., Falk, A., Huffman, D., Sunde, U., Schupp, J., and Wagner, G. G. (2011). Individual risk attitudes: Measurement, determinants, and behavioral consequences. *Journal of the European Economic Association*, 9(3):522–550.

Dohmen, T., Lehmann, H., and Pignatti, N. (2016). Time-varying individual risk attitudes over the great recession: A comparison of germany and ukraine. *Journal of Comparative Economics*, 44(1):182–200.

References II

Eckel, C. C. and Grossman, P. J. (2002). Sex differences and statistical stereotyping in attitudes toward financial risk. *Evolution and Human Behavior*, 23(4):281–295.

Gneezy, U. and Potters, J. (1997). An experiment on risk taking and evaluation periods. *The Quarterly Journal of Economics*, 112(2):631–645.

Guiso, L., Sapienza, P., and Zingales, L. (2018). Time varying risk aversion. *Journal of Financial Economics*, 128(3):403–421.

Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political analysis*, 20(1):25–46.

Hanaoka, C., Shigeoka, H., and Watanabe, Y. (2018). Do risk preferences change? evidence from the great east japan earthquake. *American Economic Journal: Applied Economics*, 10(2):298–330.

Holt, C. A. and Laury, S. K. (2002). Risk aversion and incentive effects. *American Economic Review*, 92(5):1644–1655.

References III

Jakiela, P. and Ozier, O. (2019). The impact of violence on individual risk preferences: Evidence from a natural experiment. *Review of Economics and Statistics*, 101(3):547–559.

Kettlewell, N. (2019). Risk preference dynamics around life events. *Journal of Economic Behavior & Organization*, 162:66–84.

Necker, S. and Ziegelmeyer, M. (2016). Household risk taking after the financial crisis. *The Quarterly Review of Economics and Finance*, 59:141–160.

Voors, M. J., Nillesen, E. E. M., Verwimp, P., Bulte, E. H., Lensink, R., and Soest, D. P. V. (2012). Violent conflict and behavior: a field experiment in burundi. *American Economic Review*, 102(2):941–964.

Wagner, G. G., Frick, J. R., and Schupp, J. (2007). The german socio-economic panel study (soep)-evolution, scope and enhancements.